STATISTICS · UNIT 2: SAMPLING AND EXPERIMENTATION

The Study Music Experiment

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students plan research for their own school. They select random, stratified, and cluster samples, diagnose bias in survey methods, and design an experiment testing whether music affects vocabulary recall.

They interpret a simulation of random assignment and write the scope of inference for their study.

Standards Addressed

- S-IC.A.1 Statistics as a process for making inferences about a population
- S-IC.B.3 Sample surveys, experiments, and observational studies; role of randomization
- S-IC.B.5 Use data from a randomized experiment to compare two treatments (simulation)
- S-IC.B.6 Evaluate reports based on data

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a poll: who studies with music? Part 1 sampling.
Day 2	Part 2 bias or experiment design; optional: run a small class version.
Day 3	Part 3 conclusions and the design proposal.

Materials

- Student handout and Research Fact Cards
- Random digit table or calculator randInt
- Optional: headphones, word lists, timers to run the experiment in class

How the Three Versions Differ

Level	What changes
CP2	Population vs sample, random digits, stratified counts, naming bias, and basic parts of an experiment.
CP1	Compares sampling methods, completely randomized design with blinding and confounding, and a simulation-based conclusion.
Honors	Justified sampling choice, randomized block and matched pairs designs, placebo idea, p-value from simulation, and scope of inference.

Watch For (Common Misconceptions)

- Confusing random sampling with random assignment.
- Calling a voluntary or convenience sample “random.”
- Forgetting to skip repeated labels when reading random digits.
- Claiming the result applies to all students when volunteers were used.

Extension Ideas

- Run the matched pairs experiment in class and analyze the differences.
- Use technology to repeat the re-randomization 1,000 times.
- Find a news story about a study and identify its design and scope.

Answer Key – CP2

Part 1: Who Do We Ask?

#	Task	Answer
1	What is the population? Why not just take a census?	All 1,200 students; a census takes too long and costs too much.
2	Ana suggests asking her 25 friends in the band. Why is this a bad sample?	It is a convenience sample; band members probably like music more, so the result would be biased (too high).
3	Label the volunteers 01–60. Use the random digits 19223 95034 05756 to choose 5 students. Skip numbers over 60 and repeats.	19, 22, 39, 50, 34
4	Explain how to take a stratified sample of 80 students by grade with the same fraction from each grade.	80 ÷ 1,200 = of each grade: 24 from grade 9, 20 from 10, 20 from 11, 16 from 12, chosen at random within each grade.

Part 2: Watch for Bias

#	Task	Answer
5	"Don't you agree that music helps you focus?" What kind of bias does this question create? Rewrite it.	Response bias from a leading question; "Do you study with music? Does it help, hurt, or make no difference?"
6	Surveys are emailed; only 30 of 80 reply. What kind of bias could this cause?	Nonresponse bias; the people who reply may feel more strongly than those who do not.
7	Students without school email are left out. Name this bias.	Undercoverage

Part 3: Plan the Experiment

#	Task	Answer
8	Why is this an experiment and not an observational study?	The researchers assign the treatments (music or silence) instead of just observing what students already do.
9	Name the explanatory variable, the response variable, and the treatments.	Explanatory: study condition; response: number of words recalled; treatments: silence, instrumental, lyrics.
10	Explain how to randomly assign the 60 volunteers to 3 groups of 20.	Put 60 numbered slips in a hat; the first 20 drawn get silence, the next 20 instrumental, the rest lyrics.
11	Why should the silence group be part of the experiment?	It is the control group, so we can compare each music group with no music.

Reflection

#	Prompt	Look for
R1	Why does choosing the sample at random help?	It avoids favoring any group, so the sample tends to represent the population.

#	Prompt	Look for
R2	If students who study with music get lower grades, does that prove music hurts? Explain.	No; that is an observational result, and other variables (like effort or sleep) could explain it.

Answer Key – CP1

Part 1: Sampling Plans

#	Task	Answer
1	Use the digits 19223 95034 05756 28713 to choose 6 volunteers labeled 01–60.	19, 22, 39, 50, 34, 05
2	Describe a proportional stratified random sample of 80 students by grade. Why might grade be a good stratum?	24, 20, 20, 16 chosen by SRS within each grade; study habits may differ by grade, so strata reduce variability.
3	Describe a cluster sample using homerooms. How many homerooms give about 100 students? What is one advantage and one risk?	Randomly choose 4 homerooms and survey everyone (100 students). Easier to collect; homerooms may be alike, giving more variability.
4	Name the bias in each: (a) leading question "Don't you agree music helps?"; (b) 30 of 80 reply; (c) only students with school email are reached.	(a) response bias; (b) nonresponse; (c) undercoverage

Part 2: Design the Experiment

#	Task	Answer
5	Outline a completely randomized design (a diagram is fine).	60 volunteers → random assignment → 3 groups of 20 → silence / instrumental / lyrics → compare mean words recalled.
6	Explain how the experiment uses the four principles: comparison, random assignment, control, replication.	Three treatments are compared; chance decides groups; same words, time, and room for all; 20 students per group.
7	Can the experiment be double-blind? Who could be blinded and how?	Students cannot be blind to hearing music, but the person who grades the recall test can be blind to the treatment (single-blind).
8	A student says, "Just let students choose music or silence." Explain how this creates confounding.	Students who choose music may differ (e.g., in focus or study skills), so effects of music and those traits would be mixed together.

Part 3: Results and Conclusions

#	Task	Answer
9	Find the difference in means between silence and lyrics.	18.2 − 15.6 = 2.6 words
10	The club reshuffled the 40 scores into two random groups 200 times. Only 3 reshuffles gave a difference of 2.6 or more. What does this suggest?	3 ÷ 200 = 0.015; such a big difference rarely happens by chance alone, so the result is statistically significant.
11	Can the club conclude that lyrics cause lower recall? To whom can they generalize?	Yes, cause and effect, because treatments were randomly assigned; but only to students like these volunteers, because they were not a random sample of the school.

Reflection

#	Prompt	Look for
R1	What is the difference between random sampling and random assignment?	Random sampling lets you generalize to a population; random assignment lets you conclude cause and effect.
R2	Why is the silence group important even if we mainly care about lyrics?	It is the baseline that shows whether music changes recall at all.

Answer Key – Honors

Part 1: Survey Design

#	Task	Answer
1	Compare an SRS of 80, a stratified sample by grade, and a cluster sample of homerooms. Which would you choose and why?	Stratified by grade (24, 20, 20, 16) gives representation of every grade and lower variability; cluster is cheaper but homerooms may be similar; SRS could under-sample a grade.
2	Use the random digits (all of 19223 95034 05756 28713 96409 12531) to choose 10 volunteers from 01–60. List them.	19, 22, 39, 50, 34, 05, 13, 40, 25, 31
3	Write a neutral survey question and describe two ways nonresponse or response bias could still affect results and how to reduce them.	E.g., "How often do you study with music?" Follow up with non-responders; keep surveys anonymous so students answer honestly.

Part 2: Better Experiments

#	Task	Answer
4	Design a randomized block experiment that blocks by grade level. Why block?	Within each grade, randomly assign students to the 3 treatments; compare within blocks. Blocking removes variation due to grade.
5	Design a matched pairs experiment comparing silence and lyrics using each student as their own pair.	Each student studies two different 30-word lists, once in each condition, with the order chosen by a coin flip; compare the differences.
6	Why must the order be randomized and the two word lists be equally hard?	Otherwise practice/fatigue or list difficulty would be confounded with the treatment.
7	Is a placebo possible here? Describe what could serve as one.	Not a true placebo; a "white noise" or silent-headphones condition can control for wearing headphones.

Part 3: Is the Difference Real?

#	Task	Answer
8	State the question the simulation answers, in terms of chance.	If music has no effect, how likely is a difference of 2.6 or more just from random assignment?
9	In 200 re-randomizations, 3 differences were 2.6 or more. Estimate the p-value and draw a conclusion.	≈ 0.015; this is unlikely under no effect, so there is convincing evidence lyrics lower recall.
10	Write the scope of inference: cause and effect? Generalize to whom?	Cause and effect is justified (random assignment); generalize only to similar volunteers, not the whole school (no random sample).
11	Suggest one change that would let the club generalize to the whole school.	Choose participants by a random sample of the school (e.g., stratified by grade), then randomly assign treatments.

Reflection

#	Prompt	Look for
R1	When is a matched pairs design better than a completely randomized design?	When subjects vary a lot; comparing each student to themselves removes that variation.
R2	Why can a significant result still be limited in who it applies to?	Significance comes from random assignment, but generalizing needs a random sample of the population.